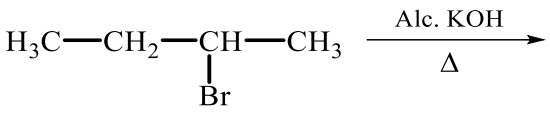
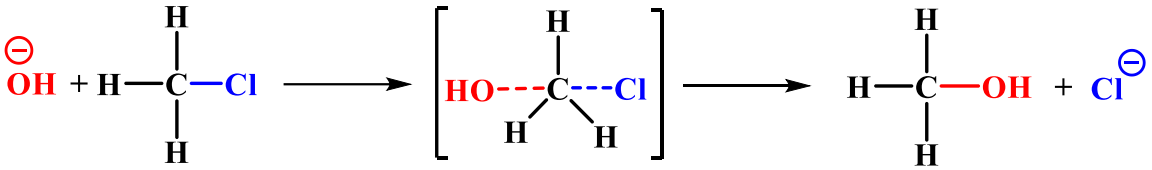
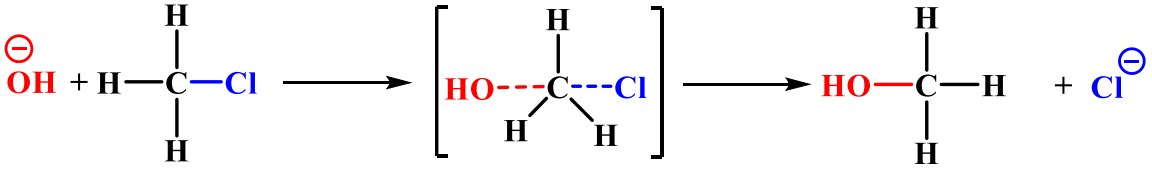
KARNATAKA SCHOOL EXAMINATION & ASSESSMENT BOARD

2026 II PUC EXAM-1

SUBJECT: **34-CHEMISTRY** SCHEME OF EVALUATION MAX. MARKS: 70

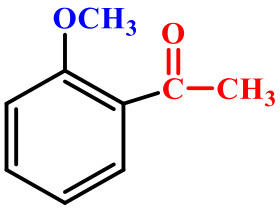
	PART-A	
I.	**Select the correct option from the given choices:**	**15 × 1=15**
1)	The major product formed in the following reaction is  a) 1-butene b) 2-butanol c) 1-bromobutane d) 2-butene	
Ans:	**d) 2-butene (or) 2-butene (or) d)**	**1**
2)	The percentage of helium filled in the tanks used by most scuba divers to dilute air in deep dives a) 32.1 b) 11.7 c) 74.2 d) 56.2	
Ans:	**b) 11.7 (or) 11.7 (or) b)**	**1**
3)	The common oxidation state shown by the element with atomic number 21 is a) +3 b) +4 c) +5 d) Both +3 and +4	
Ans:	**a) +3 (or) +3 (or) a)**	**1**
4)	The main natural source of acetic acid is a) milk b) vinegar c) red ant d) butter	
Ans:	**b) vinegar (or) vinegar (or) b)**	**1**
5)	Acetylation of salicylic acid produces a compound which has anti-inflammatory and antipyretic property. The compound is a) Bakelite b) Acetic anhydride c) Formalin d) Aspirin	
Ans:	**d) Aspirin (or) Aspirin (or) d)**	**1**
6)	p-Hydroxyazobenzene is formed by the reaction of benzene diazonium chloride with phenol. It is a) an electrophilic substitution reaction b) a nucleophilic substitution reaction c) a hydrogenation reaction d) a halogenation reaction.	
Ans:	**a) an electrophilic substitution reaction (or) an electrophilic substitution reaction (or) a)**	**1**
7)	Doctor advised a 50-year-old woman enough exposure to sunlight and addition of fish and egg yolk to her diet. Name the vitamin deficiency for the possible disease diagnosed by the doctor. a) Vitamin D b) Vitamin A c) Vitamin C d) Vitamin B_{12}	
Ans:	**a) Vitamin D (or) Vitamin D (or) a)**	**1**
8)	Which of the following statement/s is/are incorrect about order and molecularity of chemical reactions? (I) The probability that more than three molecules can collide and react simultaneously is very small. (II) There can be only three values for the order of the reaction, that is, zero, first and second. (III) The order depends on rate equation while molecularity does not. (IV) There can be infinite number of values for order. a) I and IV b) III only c) II and IV d) II only	
Ans:	**d) II only (or) II only (or) d)**	**1**

9)	Which of the following groups when present at para position increases the basic strength of aniline? a)-NO_2 b) -Br c) -NH_2 d) -COOH	
Ans:	**c) -NH_2 (or) -NH_2 (or) c)**	**1**
10)	Among the following cells, the cell used in the apollo space program for providing electric power is a) SHE b) H_2-O_2 fuel cell c) Daniel cell d) Mercury cell	
Ans:	**b) H_2-O_2 fuel cell (or) H_2-O_2 fuel cell (or) b)**	**1**
11)	Isopropyl magnesium bromide and n-propyl magnesium bromide react separately with H_2O to yield the organic compounds, respectively are a) Isopropyl bromide and n-propyl bromide b) Isopropyl alcohol and n-propyl alcohol c) n-propyl alcohol and isopropyl alcohol d) Propane and propane	
Ans:	**d) Propane and propane (or) Propane and propane (or) d)**	**1**
12)	A binary solution has two components 'A' and 'B'. The mole fraction of component 'A' is 0.5, then the number of moles of components 'A and 'B' in the solution is a) $n_A > n_B$ b) $n_A < n_B$ c) $n_A = n_B$ d) zero	
Ans:	**c)** $n_A = n_B$ **(or)** $n_A = n_B$ **(or) c)**	**1**
13)	Match the following given in List I with List II List I (Elements) / List II (Their maximum oxidation states) i) Thorium (Th) — a) +7 ii) Protactinium (Pa) — b) +6 iii) Lawrencium (Lr) — c) +3 iv) Uranium (U) — d) +5 v) Plutonium (Pu) — e) +4 Choose the correct option. a) i-e, ii-d, iii-a, iv-b, v-c b) i-e, ii-d, iii-c, iv-b, v-a c) i-b, ii-a, iii-c, iv-e, v-d d) i-b, ii-e, iii-c, iv-d, v-a	
Ans:	**b) i-e, ii-d, iii-c, iv-b, v-a (or) i-e, ii-d, iii-c, iv-b, v-a (or) b)**	**1**
14)	The complex $\left[Co(NH_3)_5(NO_2)\right]Cl_2$ is obtained in the red form when nitrite ligand is bound to cobalt through oxygen (–ONO) and its linkage isomer is obtained as a _____ form when nitrite ligand is bound to "Co" through nitrogen (–NO_2). a) orange b) blue c) white d) yellow	
Ans:	**d) yellow (or) yellow (or) d)**	**1**
15)	Given below are two statements: **Statement I:** Alcohols are acidic in nature; The acidic character of alcohols is due to the polar nature of the O–H bond in it. **Statement II:** Alcohols are weaker acids than water. In the light of the above statements, choose the most appropriate answer from the options given below: a) Statement I is incorrect but Statement II is correct. b) Both Statement I and Statement II are correct. c) Both Statement I and Statement II are incorrect. d) Statement I is correct but Statement II is incorrect.	
Ans:	**b) Both Statement I and Statement II are correct.** **(or) Both Statement I and Statement II are correct. (or) b)**	**1**

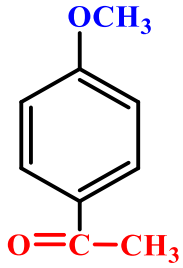
II.	**Fill in the blanks by choosing the appropriate word from those given in the brackets: [reducing, zero, sodium alkoxide, one, pleasant, pungent]**	**5×1=5**
16)	Oxidation State of 'Ni' in $[Ni(CO)_4]$ is _____.	
Ans:	**zero**	**1**
17)	In Williamson ether synthesis, an alkyl halide is allowed to react with _____.	
Ans:	**sodium alkoxide**	**1**
18)	The carbohydrates which reduce Fehling's solution and Tollens' reagent are called as _____ sugars.	
Ans:	**reducing**	**1**
19)	The lower aldehydes have sharp _____ odours.	
Ans:	**pungent**	**1**
20)	Van't Hoff factor (i) for a non-electrolyte in a solution is _____.	
Ans:	**one**	**1**
	PART - B	
III.	**Answer any three of the following. Each question carries 2 marks.**	**3 × 2 = 6**
21)	**What are transition elements? Give an example.**	
Ans:	Elements which have incomplete *d* subshell (partially filled *d*-orbitals) either in neutral atom (or) in their ions are called transition elements. **(or)** The d-block elements, which are having incompletely filled (n-1)d orbitals in their atomic (or) ionic states are called transition elements. **Example:** Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu and etc., **(Any suitable example 1 mark)**	**1** **1**
22)	**Write the S_N2 mechanism involved in the conversion of chloromethane to methanol.**	
Ans:	 **(or)**  **(1 mark for each step)**	**2** **(or)** **2**
23)	**What is half-life of a reaction? Represent it symbolically.**	
Ans:	The half-life of a reaction is the time in which the concentration of a reactant is reduced to one half of its initial concentration. It is represented as "$t_{½}$" **(or)** "$t_{50\,\%}$" **(or)** "$t_{0.5}$"	**1** **1**
24)	**List any two nitrogenous bases commonly found in both DNA and RNA.**	
Ans:	Adenine, Guanine and Cytosine **(or)** (A), (G) and (C) **(Any two; for each 1 mark)**	**2**

25) **Write the products formed when anisole reacts with ethanoyl chloride in the presence of anhydrous $AlCl_3$ catalyst.**

Ans:



and



(or)

2-methoxyacetophenone
(or)
o-methoxy acetophenone
and
4-methoxyacetophenone
(or)
p-methoxy acetophenone

1+1

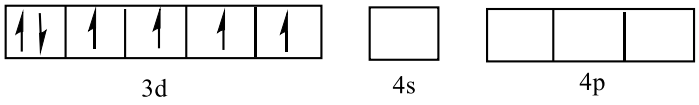
PART - C

IV. **Answer any three of the following. Each question carries 3 marks.** **3 × 3 = 9**

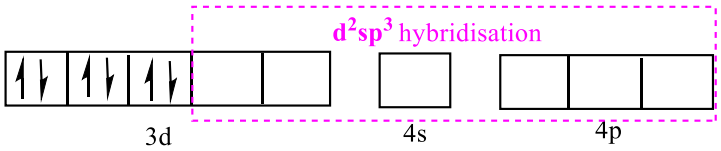
26) **Using Valence Bond Theory: account for hybridisation, geometry and magnetic property of $[Co(NH_3)_6]^{3+}$ ion.**
[Given: Atomic mass of cobalt is 27].

Ans:

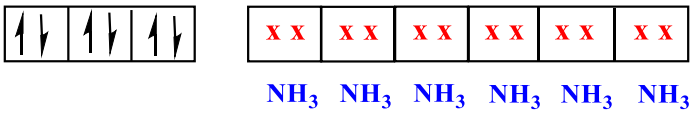
Electronic configuration of Co^{3+}: $[Ar]\ 3d^6\ 4s^0$



In presence of NH_3 electrons pairing takes place;



$[Co(NH_3)_6]^{3+}$:



Hybridisation: d^2sp^3.

Magnetic property: Diamagnetic.

Geometry: Octahedral.

Geometry: one-mark, Magnetic property: one mark; Explanation: one mark

27) **What is Lanthanoid contraction? Mention two consequence of it.**

Ans:

The overall decrease in atomic radii and ionic radii from lanthanum to lutetium (across lanthanoids) is called Lanthanoid contraction. **1**

1. The separation of lanthanoids in pure state becomes difficult. **(or)**

   Difficulty in separation of lanthanoids due to similar chemical properties.

2. The atomic radii of 3rd row transition series elements are almost similar to that of 2nd row transition series elements. **(or)**

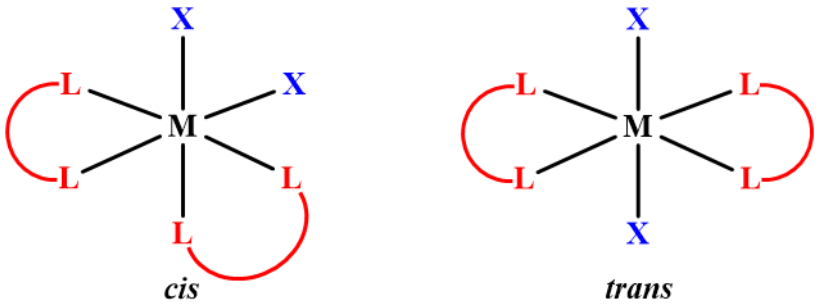
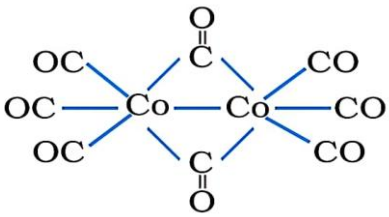
   The identical radii of Zirconium (Zr) and Hafnium (Hf).

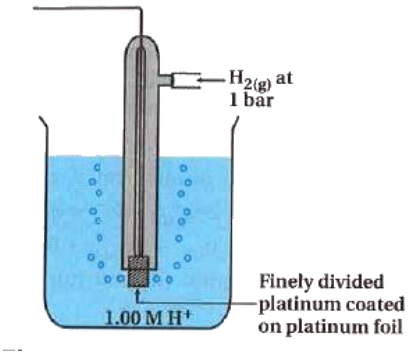
3. The covalent character increases.

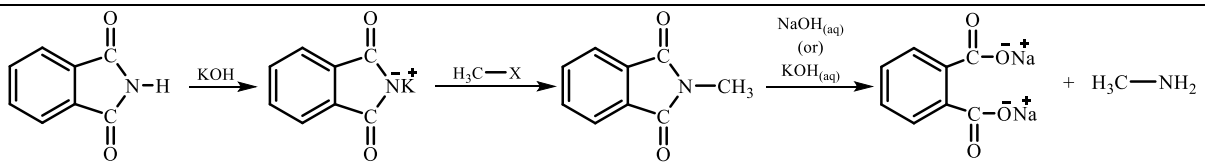
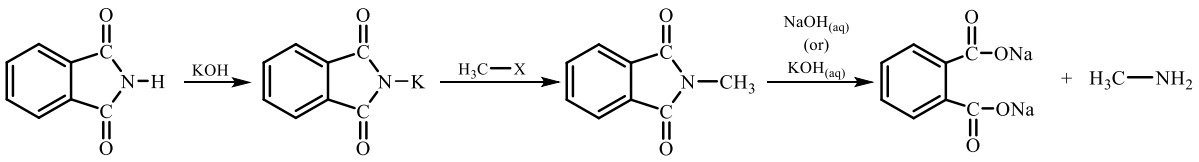
4. Basicity decreases.

2

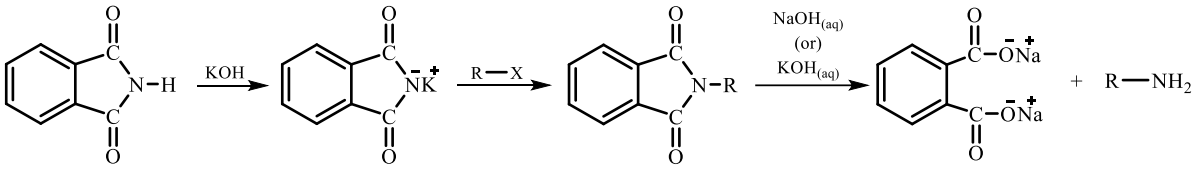
(Any correct consequences: 1 mark for each)

28)	**Write the geometrical isomers of the complex $[MX_2(L-L)_2]$, where "M", "X" and "L-L" are central metal atom/ion, monodentate ligand and didentate ligand respectively. Identify the optically inactive form of it.**	
Ans:	 Trans form is optically inactive.	**1+1** **1**
29)	**Complete the following chemical equations.** **(i) _____ + 8 Na_2CO_3 + 7 O_2 → 8 Na_2CrO_4 + 2 Fe_2O_3 + 8 CO_2** **(ii) 2 Na_2CrO_4 + _____ → $Na_2Cr_2O_7$ + 2 Na^+ + H_2O** **(iii) $Na_2Cr_2O_7$ + 2 KCl → _____ + 2 NaCl**	
Ans:	(i) 4 $FeCr_2O_4$ **(or)** 4 $FeO.Cr_2O_3$ (ii) 2 H^+ **(Strictly H_2SO_4 and 2HCl are not to be considered as an answer)** (iii) $K_2Cr_2O_7$	**1** **1** **1**
30)	**Write the IUPAC name and structure of $[Co_2(CO)_8]$ and mention the number of bridged CO groups.**	
Ans:	Octacarbonyldicobalt(0)  Two (or) 2	**1** **1** **1**
V	**Answer any two of the folowing. Each question carries 3 marks:**	**2 × 3 = 6**
31)	**Derive an integrate rate equation for the rate constant of first-order reaction.**	
Ans:	Consider the general first order reaction R → Products Rate = $\frac{-d[R]}{dt} = k[R]$ (or) $\frac{d[R]}{[R]} = -k.dt$ Integrate on both sides, we get $\ln [R] = -kt + I$ When, $t = 0$, then $R = [R]_0$, where $[R]_0$ is the initial concentration of the reactant.	**1**

	$\ln [R]_0 = - k \times 0 + I$ $\ln [R]_0 = I$ Substitute the value of I in the above equation. $\ln [R] = - k\, t + \ln [R]_0$ Rearrange, $\ln \frac{[R]}{[R]_0} = -k\,t$ $k = \frac{1}{t} \ln \frac{[R]_0}{[R]}$ **(or)** $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$	**1** **1**
32)	**Write a neat labelled diagram, cell representation and half-cell reaction of Standard Hydrogen Electrode (S.H.E.).**	
Ans:	 **(Any two labelling 1 mark)** **Cell representation:** $Pt(s)\,|H_2(g)\,|H^+(aq)$ **Half-cell reaction:** **As anode:** $\frac{1}{2}H_{2(g)} \rightarrow H^+_{(aq)} + e^-$ **(or)** **(or)** $H^+_{(aq)} + e^- \rightleftharpoons \frac{1}{2}H_{2(g)}$ **As cathode:** $H^+_{(aq)} + e^- \rightarrow \frac{1}{2}H_{2(g)}$ **(Any one cell reaction; 1 mark)**	**1** **1** **1**
33)	**What are ideal solutions? Write two characteristics of it.**	
Ans:	The solutions which obey Raoult's law over the entire range of concentration are known as ideal solutions. **Characteristics:** 1. The enthalpy of mixing of the pure components to form the solution is zero. **(or)** $\Delta_{mix}H = 0$ 2. The volume of mixing is also zero. **(or)** $\Delta_{mix}V = 0$ 3. Intermolecular attractive forces between A-A, B-B and A-B are same. 4. $P_{total} = P^o_A\chi_A + P^o_B\chi_B$ (or) $P_{total} = P_A + P_B$ **(Any two characteristics; 1 mark each)**	**1** **2**
34)	**Mention any three factors which decide the products of electrolysis.**	
Ans:	1. Nature of material being electrolysed. 2. Type of electrodes. 3. It depend on the different oxidising and reducing species present in the electrolytic cell and their standard electrode potentials. 4. Concentration of electrolytic solution. **(Any three factors, 1 mark each)**	**3**

	PART - D	
VI.	**Answer any four of the following. Each question carries 5 marks.**	**4 × 5 = 20**
35) a)	**Name the two monosaccharides obtained when lactose is subjected to hydrolysis.**	
b)	**What are non-essential amino acids? Mention the name of a α-amino acid which is optically inactive.**	**(2+2+1)**
c)	**Name the hormone which mediates responses to external stimuli.**	
Ans: a)	β-D-galactose and β-D-glucose **(Strictly $\beta-(D)$ forms are expected)**	**1+1**
b)	The amino acids, which **can be synthesised in the body**, are known as non-essential amino acids.	**1**
	Glycine (or) Gly	**1**
c)	Epinephrine (Adrenaline) **(or)** Norepinephrine (Noradrenaline) **(Any one hormone; 1 mark)**	**1**
36) a)	**When an alkyl chloride reacts with sodium iodide in dry acetone gives alkyl iodide.** **(i) Name this reaction.** **(ii) Write the general equation.** **(iii) Mention the role of dry acetone.**	**(3+2)**
b)	**What is meant by racemic modification? "They are optically inactive". Give reason.**	
Ans: a)	(i) Finkelstein reaction.	**1**
	(ii) $RX + NaI \xrightarrow{\text{Dry Acetone}} RI + NaX$	**1**
	(iii) Driving the reaction forward by promoting precipitation of the by-product sodium halide (NaX). **(or)** Dry acetone facilitates the forward reaction. (**or)** Serves as a solvent. **(or)** Facilitating the S_N2 reaction by dissolving reactants.	**1**
b)	A mixture containing two enantiomers in equal proportions will have zero optical rotation is called racemic modification (racemic mixture). **(or)** Equimolar mixture of *d* and *l* isomers is called racemic modification (racemic mixture). **(or)** Mixture of 50% *d* and 50% *l* isomers is called racemic modification (racemic mixture).	**1**
	Because the optical rotation due to one isomer will be cancelled by the rotation due to the other isomer.	**1**
37) a)	**Write the chemical equations involved in the preparation of methanamine through Gabriel phthalimide synthesis.**	**(3+2)**
b)	**Illustrate the carbylamine reaction with general chemical equation.**	
Ans: a)		**3**
	(or)	**(or)**
		**3**

(or) **(or)**

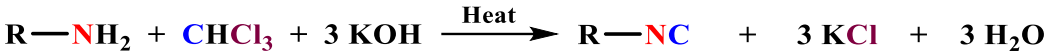


3

Where R is -CH_3

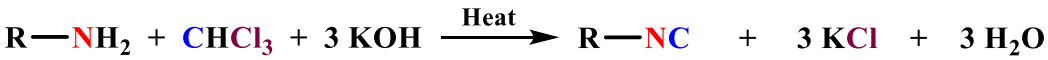
(1 mark for each step)

b) Aliphatic and aromatic primary amines on heating with chloroform and ethanolic potassium hydroxide form isocyanides or carbylamines. This reaction is known as carbylamine reaction. **1**



1

(or) **(or)**

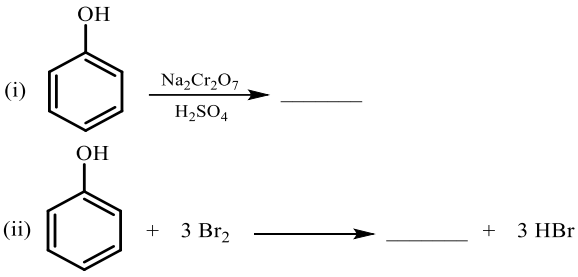


2

1°-amine **chloroform** **Carbylamine**

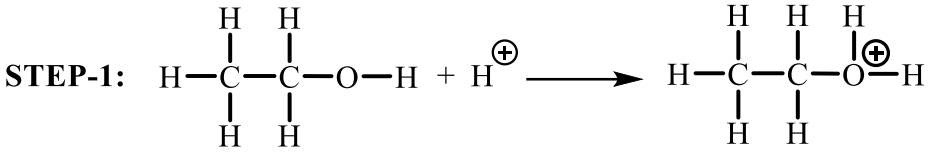
38) a) **Write the steps involved in the mechanism of acid catalysed dehydration of ethanol to ethene.**

b) **Complete the following reactions;**

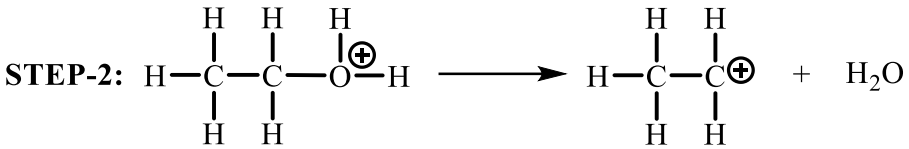


(3+2)

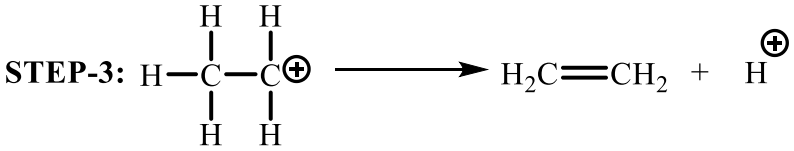
Ans: a)



1



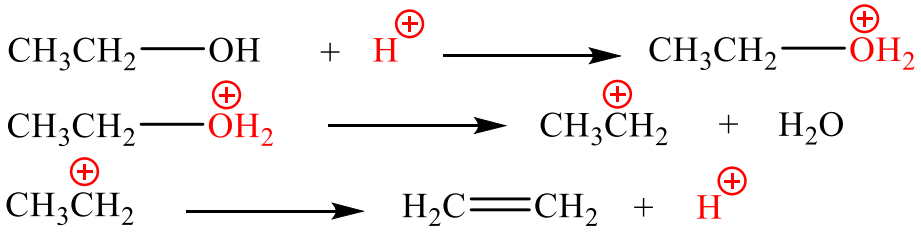
1



1

(In this mechanism, charges are strictly expected where it is necessary)

(or) **(or)**

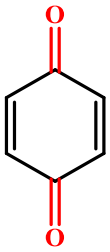
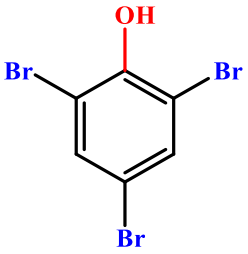


1

1

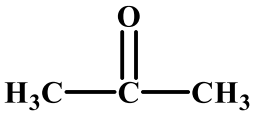
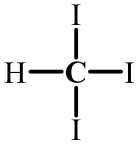
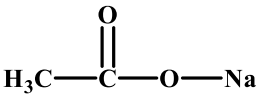
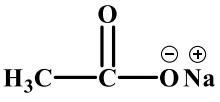
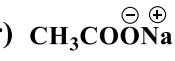
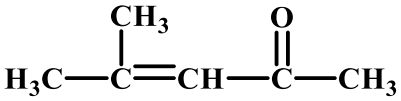
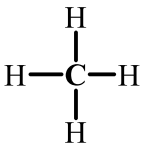
1

(In this mechanism, charges are strictly expected where it is necessary)

b) (i)  (ii)  **1+1**

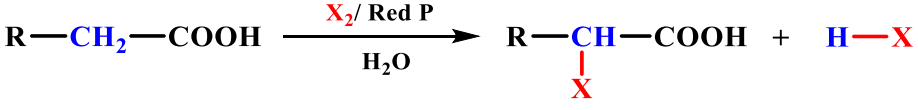
39) Addition of H_2O to an organic compound with chemical formula C_3H_4 in the presence of sulphuric acid and mercury (II) sulphate gives "A". "A" is oxidized with sodium hypoiodite gives "B" and "C". When "A" is heated with barium hydroxide (catalyst) forms "D" by readily loss of water. "C", upon heating with sodium hydroxide and calcium oxide in the ratio of 3:1 gives an organic compound "E". Write the structures of "A", "B", "C", "D" and "E".

Ans:

'A'	 **(or)** **H_3CCOCH_3**	**1**
'B'	 (or) **CHI_3**	**1**
'C'	 **(or)**  **(or)** **CH_3COONa** **(or)** 	**1**
'D'	 **(or)** **$(CH_3)_2C=CH-CO-CH_3$**	**1**
'E'	 (or) **CH_4**	**1**

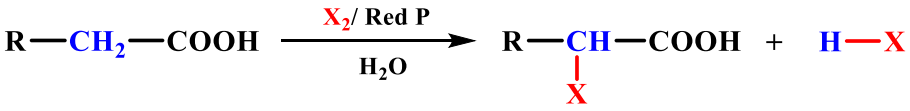
40) a) Explain Hell-Volhard–Zelinsky reaction with chemical equation.
b) Write the chemical reaction for the conversion of toluene to benzaldehyde using chromyl chloride. Name the reaction.
c) Name the gas used to facilitate the nucleophilic addition of ethylene glycol to aldehydes and ketones. **(2+2+1)**

Ans: a) Carboxylic acids having an α-hydrogen are halogenated at the α -position on treatment with halogen in the presence red phosphorus to give α-halocarboxylic acids. This reaction is known as Hell-Volhard-Zelinsky reaction. **1**



1

(or)

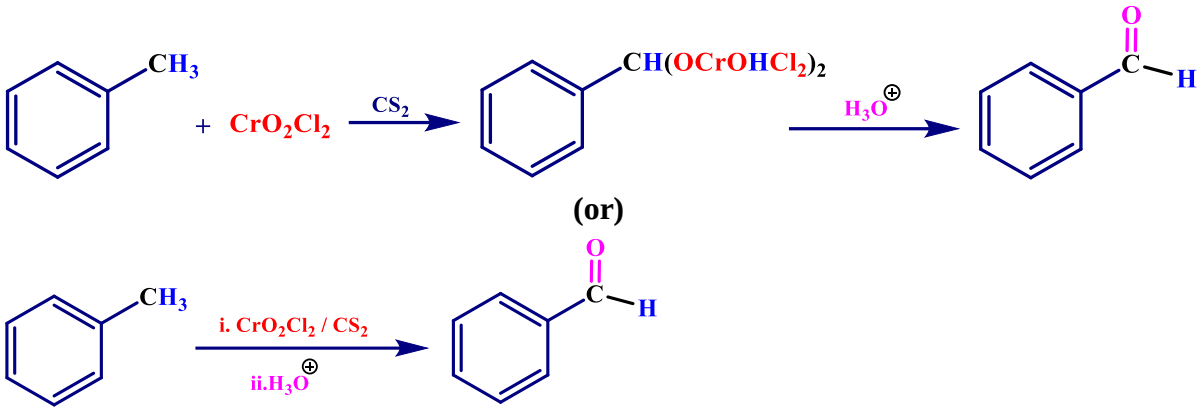


2

Carboxylic acid α-halo carboxylic acid

(Any suitable example with self-explanatory equation 2 mark)

b)



(or)

1

Etard reaction **1**

c) Dry HCl gas **(or)** Dry Hydrogen chloride gas **1**

PART – E (PROBLEMS)

VII. **Answer any three of the following. Each question carries 3 marks.** **3 × 3 = 9**

41) **Calculate the mass of a non-volatile solute which should be dissolved in 114g octane to reduce its vapour pressure to 80%. [Given Molecular formula of octane is C_8H_{18}; Molar mass of solute = 40 g mol^{-1}]**

Ans:

For a dilute solution:

$$\frac{p_1^o - p_1}{p_1^o} = \frac{w_2 \times M_1}{M_2 \times w_1}$$ **1**

$$\frac{100-80}{100} = \frac{w_2 \times 114}{40 \times 114} \quad \text{(or)} \quad \frac{100-80}{100} = \frac{w_2}{40} \times 1$$ **1**

$$\frac{20}{100} = \frac{w_2}{40} \Rightarrow 5w_2 = 40 \Rightarrow w_2 = \frac{40}{5}$$

$$w_2 = 8.0\,g$$ **1**

(or) **(or)**

For a concentrated solution:

$$\frac{p_1^o - p_1}{p_1^o} = \frac{n_2}{n_1 + n_2}$$ **1**

$$\frac{100-80}{100} = \frac{\frac{w_2}{40}}{\frac{w_2}{40} + \frac{114}{114}} \quad \text{(or)} \quad \frac{20}{100} = \frac{\frac{w_2}{40}}{\frac{w_2}{40} + 1}$$ **1**

$$\frac{20}{100} = \frac{w_2}{40} \times \frac{40}{w_2 + 40} \Rightarrow 5w_2 = w_2 + 40 \Rightarrow 5w_2 - w_2 = 40 \Rightarrow 4w_2 = 40$$

$$w_2 = 10.0\,g$$ **1**

(Strictly expecting correct unit with final answer)

42) **The rate constant for the first order decomposition of H_2O_2 is given by the following equation: $\log k = 14.34 - 1.25 \times 10^4\ K/T$. Calculate E_a for the reaction. [Given: R= 8.314 JK^{-1}mol^{-1}]**

Ans:	Given; $\log k = 14.34 - 1.25\times10^4 \frac{K}{T}$ - - - - - - - - - (1) We know that: $\log k = \log A - \frac{E_a}{2.303RT}$ - - - - - - - -(2) On comparing equation (1) and (2) $\frac{E_a}{2.303RT} = \frac{1.25\times10^4}{T}$ $E_a = 1.25\times10^4 \times 2.303 \times 8.314$ $E_a = 239339.275\ J mol^{-1}$ (or) $E_a = 239.339\ kJ mol^{-1}$ (or) $E_a = 239.34\ kJ mol^{-1}$ **(Strictly expecting correct unit with final answer)**	**1** **1** **1**
43)	**The cell in which the reaction occurs** $2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \rightleftharpoons 2Fe^{2+}_{(aq)} + I_{2(s)}$; $E^o_{cell} = 0.230\,V$ at $298\,K$ **Calculate the value of log K_c (K_c = equilibrium constant) of the cell reaction.**	
Ans:	$E^o_{Cell} = \frac{0.0591}{n} \times \log K_c$ (or) $\log K_c = \frac{E^o_{Cell} \times n}{0.0591}$ [OR] $E^o_{Cell} = \frac{0.059}{n} \times \log K_c$ (or) $\log K_c = \frac{E^o_{Cell} \times n}{0.059}$ $\log K_c = \frac{0.230\times2}{0.0591}$ [OR] $\log K_c = \frac{0.230\times2}{0.059}$ $\log K_c = 7.7834$ (or) $\log K_c = 7.78$ [OR] $\log K_c = 7.7966$ (or) $\log K_c = 7.79$	**1** **1** **1**
44)	**1.00 g of a non-electrolyte solute dissolved in 50 g of benzene lowered the freezing point of benzene by 0.40 K. The freezing point depression constant of benzene is 5.12 K kg mol^{-1}. Find the molar mass of the solute.**	
Ans:	$M_2 = \frac{1000\times w_2 \times K_f}{\Delta T_f \times w_1}$ (or) $M_B = \frac{1000\times w_B \times K_f}{\Delta T_f \times w_A}$ $M_2 = \frac{1000\times1\times5.12}{0.4\times50}$ $M_2 = 256\,g\,mol^{-1}$ **(Strictly expecting correct unit with final answer)**	**1** **1** **1**
45)	**Calculate the limiting molar conductivity of Cl^- ion by using the data: $\lambda_{Ca^{2+}} = 119.0\ Scm^2mol^{-1}$ and Λ^o_m for $CaCl_2$ = 271.6 Scm^2mol^{-1}.**	
Ans:	$\Lambda^o_{mCaCl_2} = \lambda^o_{mCa^{2+}} + 2\lambda^o_{mCl^-}$ (or) $2\lambda^o_{mCl^-} = \Lambda^o_{mCaCl_2} - \lambda^o_{mCa^{2+}}$ $\lambda^o_{mCl^-} = \frac{271.6 - 119.0}{2}$ $\lambda^o_{mCl^-} = 76.3 Scm^2mol^{-1}$ **(Strictly expecting correct unit with final answer)**	**1** **1** **1**

46)	**A first order reaction takes 40 min for 30% decomposition. Calculate $t_{½}$ (half-life period).**	
Ans:	$k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$	**1**
	$k = \frac{2.303}{40}\log\frac{100}{70}$ (or) $k = \frac{2.303}{40\times60}\log\frac{100}{70}$	
	$k = \frac{2.303}{40}\times0.1549$ $k = \frac{2.303}{40\times60}\times0.1549$	
	$k = 0.00891\,\text{min}^{-1}$ (or) $k = 8.91\times10^{-3}\,\text{min}^{-1}$ (or) $k = 0.0001486\,\text{s}^{-1}$ (or) $k = 1.486\times10^{-4}\,\text{s}^{-1}$	**1**
	$t_{½} = \frac{0.693}{k} \Rightarrow t_{½} = \frac{0.693}{0.00891} \Rightarrow t_{½} = 77.77\,\text{min}$ (or) $t_{½} = \frac{0.693}{k} \Rightarrow t_{½} = \frac{0.693}{0.0001486} \Rightarrow t_{½} = 4663.5\,\text{s}$	**1**
	(Strictly expecting correct unit with final answer)	

(This is a final scheme of evaluation given by KSEAB. There is no scope to change)

*********END*********